

EXPERIMENT-7

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Q1 Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

AB→C, C→D, D→A

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Sol:

Given:

Relation: $R(A, B, C, D)$

Functional Dependencies:

1. $AB \rightarrow C$
2. $C \rightarrow D$
3. $D \rightarrow A$

1. AB^+

- $AB \rightarrow C \Rightarrow$ get C
- $C \rightarrow D \Rightarrow$ get D
- $D \rightarrow A \Rightarrow$ already have A

So:

$$AB^+ = \{A, B, C, D\}$$

Therefore, AB is a superkey.

Since neither A nor B alone determines all attributes, AB is a candidate key.

2. BC^+

- $C \rightarrow D \Rightarrow$ get D
- $D \rightarrow A \Rightarrow$ get A
- Now have A and B, so $AB \rightarrow C$, already have C.

Thus:

$$BC^+ = \{A, B, C, D\}$$

So BC is a candidate key.

3. BD^+

- $D \rightarrow A \Rightarrow$ get A
- Now have AB
- $AB \rightarrow C \Rightarrow$ get C

Thus:

$$BD^+ = \{A, B, C, D\}$$

So BD is a candidate key.

Candidate Keys

$$\{AB, BC, BD\}$$

Step 2: Prime Attributes

A prime attribute is an attribute that is part of at least one candidate key.

$$\text{Prime attributes} = \{A, B, C, D\}$$

Q2 Relation R(ABCDE) having functional dependencies as:

A $\rightarrow$ D, B $\rightarrow$ A, BC $\rightarrow$ D, AC $\rightarrow$ BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Sol:

Given:

Relation: R(A,B,C,D,E)

Functional Dependencies: A $\rightarrow$ D, B $\rightarrow$ A, BC $\rightarrow$ D, AC $\rightarrow$ BE

Using closure:

1. (AC)⁺

AC $\rightarrow$ B,E, so add B and E.

A $\rightarrow$ D, so add D.

Therefore, (AC)⁺ = ABCDE.

AC is a candidate key because neither A nor C alone determines all attributes.

2. (BC)⁺

B $\rightarrow$ A, so add A.

A $\rightarrow$ D, so add D.

AC $\rightarrow$ BE, and B and C are already present.

Therefore, (BC)⁺ = ABCDE.

BC is a candidate key because neither B nor C alone determines all attributes.

Candidate Keys = {AC, BC}

Prime Attributes = {A, B, C}

Non-Prime Attributes = {D, E}

Q3 Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

$B \rightarrow A, A \rightarrow C, BC \rightarrow D, AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Sol:

Given:

Relation: R(A,B,C,D,E)

Functional Dependencies: $B \rightarrow A, A \rightarrow C, BC \rightarrow D, AC \rightarrow BE$

Using closure:

1. B^+

$B \rightarrow A$, so add A.

$A \rightarrow C$, so add C.

$BC \rightarrow D$, so add D.

$AC \rightarrow BE$, so add E.

Therefore, $B^+ = ABCDE$.

B is a candidate key.

2. A^+

$A \rightarrow C$, so add C.

$AC \rightarrow BE$, so add B and E.

$BC \rightarrow D$, so add D.

Therefore, $A^+ = ABCDE$.

A is a candidate key.

Candidate Keys = {A, B}

Prime Attributes = {A, B}

Non-Prime Attributes = {C, D, E}

Q4 Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

$A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Sol:

Given:

Relation: R(A,B,C,D,E,F)

Functional Dependencies: $A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A$

Since F never appears on the RHS of any FD, F must be present in every candidate key.

Using closure:

1. (AF)⁺

A → B,C,D, so add B, C and D.

BC → D,E, so add E.

Therefore, (AF)⁺ = ABCDEF.

AF is a candidate key.

2. (BF)⁺

B → D, so add D.

D → A, so add A.

A → B,C,D, so add C.

BC → D,E, so add E.

Therefore, (BF)⁺ = ABCDEF.

BF is a candidate key.

3. (DF)⁺

D → A, so add A.

A → B,C,D, so add B and C.

BC → D,E, so add E.

Therefore, (DF)⁺ = ABCDEF.

DF is a candidate key.

Candidate Keys = {AF, BF, DF}

Prime Attributes = {A, B, D, F}

Non-Prime Attributes = {C, E}

Q5 Relation R(ABCDE) having dependencies as:

CE → D

D → B

C → A

Highest normal form? C.K set?

Sol:

Candidate key: CE

CE⁺ = C,E → D → B and C → A, hence CE⁺ = ABCDE.

Neither C nor E alone determines all attributes, so CE is minimal.

Prime Attributes = {C, E}; Non-Prime Attributes = {A, B, D}.

C → A is a partial dependency because C is a proper subset of candidate key CE and A is non-prime.

Therefore, the relation violates 2NF. Assuming atomic values (1NF), the highest

normal form is 1NF.

Q6 Relation R(ABCDEF) having dependencies as:

AB $\rightarrow$ C

DC $\rightarrow$ AE

E $\rightarrow$ F

Highest normal form? C.K set?

Sol:

Candidate keys: ABD and BCD.

ABD⁺ = A,B,D $\rightarrow$ C; then DC $\rightarrow$ A,E; and E $\rightarrow$ F. Hence ABD⁺ = ABCDEF.

BCD⁺ = B,C,D $\rightarrow$ A,E; and E $\rightarrow$ F. Hence BCD⁺ = ABCDEF.

Prime Attributes = {A, B, C, D}; Non-Prime Attributes = {E, F}.

DC $\rightarrow$ E is a partial dependency because DC is a proper subset of candidate key BCD and E is non-prime.

Therefore, the relation is not in 2NF. Assuming atomic values, highest normal form = 1NF.

Q7 Relation R(ABCDEFGHI) having dependencies as:

AB $\rightarrow$ C

BD $\rightarrow$ EF

AD $\rightarrow$ GH

A $\rightarrow$ I

Highest normal form? C.K set?

Sol:

Candidate key: ABD.

ABD⁺ = A,B,D $\rightarrow$ C,E,F,G,H and A $\rightarrow$ I. Hence ABD⁺ = ABCDEFGHI.

Since A, B and D do not all occur on the RHS, they are required for the key.

Prime Attributes = {A, B, D}; Non-Prime Attributes = {C, E, F, G, H, I}.

AB $\rightarrow$ C is a partial dependency because AB is a proper subset of candidate key ABD and C is non-prime.

Therefore, the relation is not in 2NF. Assuming atomic values, **highest normal form = 1NF.**

Q8 Relation R(ABCDE) having dependencies as:

AB $\rightarrow$ CD

D $\rightarrow$ A

BC $\rightarrow$ DE

Highest normal form? C.K set?

Sol:

Candidate keys: AB and BC.

$AB^+ = A, B \rightarrow C, D$; then $BC \rightarrow D, E$. Hence $AB^+ = ABCDE$.

$BC^+ = B, C \rightarrow D, E$; then $D \rightarrow A$. Hence $BC^+ = ABCDE$.

Prime Attributes = {A, B, C}; Non-Prime Attributes = {D, E}.

The relation is in 2NF because there is no dependency from a proper subset of a candidate key to a non-prime attribute.

$D \rightarrow A$ violates 3NF because D is not a superkey, while A is prime.

Therefore, **highest normal form = 2NF.**

Q9 Relation R(ABCDE) having dependencies as:

$BC \rightarrow ADE$

$D \rightarrow B$

Highest normal form? C.K set?

Sol:

Candidate keys: BC and CD.

$BC^+ = A, B, C, D, E$ directly from $BC \rightarrow ADE$, so BC is a candidate key.

$CD^+ = C, D \rightarrow B$ through $D \rightarrow B$; then $BC \rightarrow A, D, E$. Hence $CD^+ = ABCDE$, so CD is a candidate key.

Prime Attributes = {B, C, D}; Non-Prime Attributes = {A, E}.

$D \rightarrow B$ does not violate 3NF because B is a prime attribute, although D is not a superkey.

However, $D \rightarrow B$ violates BCNF because D is not a superkey.

Therefore, **highest normal form = 3NF.**

Q10 Relation R(ABC) having dependencies as:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Highest normal form? C.K set?

Sol:

$A^+ = ABC$, so A is a candidate key.

$B^+ = ABC$, so B is a candidate key.

$C^+ = ABC$, so C is a candidate key.

Candidate Keys = {A, B, C}.

Prime Attributes = {A, B, C}; Non-Prime Attributes = none.

For every FD $A \rightarrow B$, $B \rightarrow C$ and $C \rightarrow A$, the determinant is a candidate key/superkey.

Therefore, the relation satisfies BCNF.

Highest normal form = BCNF.

Q11 Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 → F3

F2 → F4

(F1, F2) → F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Sol:

Given: P(F1, F2, F3, F4, F5)

Functional Dependencies: F1 → F3, F2 → F4, F1F2 → F5

Candidate Key: (F1, F2), because (F1F2)⁺ = F1F2F3F4F5.

Prime Attributes = {F1, F2}

Non-Prime Attributes = {F3, F4, F5}

F1 → F3 and F2 → F4 are partial dependencies because F1 and F2 are proper subsets of the candidate key and F3 and F4 are non-prime attributes.

Therefore, the relation violates 2NF. Assuming atomic values, **the highest normal form is 1NF.**

Q12 Consider a relation R(ABCDEFGHI) having FDs as:

AB → C

AD → GH

BD → EF

A → I

H → J

Identify that relation is in which normal form and the candidate key by determining the prime and non prime attributes.

Apply the decomposition if needed to remove the redundancy.

Identify how many new tables will be formed after the decomposition.

Sol:

Candidate Key: ABD

ABD⁺ = A, B, D → C, G, H, E, F; A → I; H → J. Hence ABD⁺ = ABCDEFGHIJ.

Prime Attributes = {A, B, D}

Non-Prime Attributes = {C, E, F, G, H, I, J}

AB → C, AD → GH, BD → EF and A → I are partial dependencies because their determinants are proper subsets of candidate key ABD and their dependent attributes are non-prime.

Therefore, the relation is not in 2NF. Assuming atomic values, **the highest normal form is 1NF.**

Q13 Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows:

A $\rightarrow$ B

B $\rightarrow$ E

C $\rightarrow$ D

Identify that relation is in which normal form and the candidate key by determining the prime and non prime attributes.

Apply the decomposition if needed to remove the redundancy.

Identify how many new tables will be formed after the decomposition.

Sol:

Candidate Key: AC

$AC^+ = A, C \rightarrow B, D; B \rightarrow E$. Hence $AC^+ = ABCDE$.

Prime Attributes = {A, C}

Non-Prime Attributes = {B, D, E}

A $\rightarrow$ B and C $\rightarrow$ D are partial dependencies because A and C are proper subsets of candidate key AC and B and D are non-prime.

Therefore, the relation is not in 2NF. Assuming atomic values, **highest normal form = 1NF.**

B $\rightarrow$ E is also a transitive dependency, so the relation is not in 3NF.